

Problem 105 (Czech-Polish-Slovak Match 2013 Problem 2). Let x be a positive real number, and let n be a positive integer. Prove that

$$x^n + \frac{1}{x^n} - 2 \geq n^2 \left(x + \frac{1}{x} - 2 \right).$$

Solution. Note that

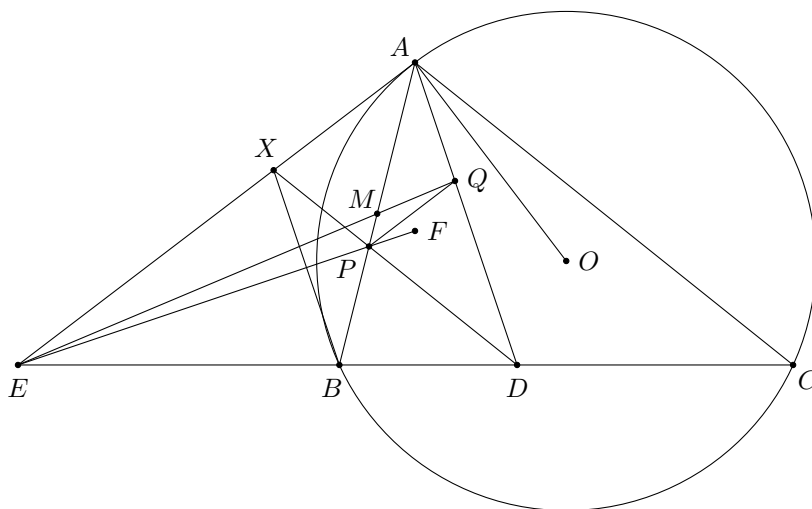
$$\begin{aligned} & x^n + \frac{1}{x^n} - 2 \geq n^2 \left(x + \frac{1}{x} - 2 \right) \\ \Leftrightarrow & x^{2n} - 2x^n + 1 \geq n^2 x^{n-1} (x^2 - 2x + 1) \\ \Leftrightarrow & (x-1)^2 (x^{n-1} + x^{n-2} + \dots + 1)^2 \geq n^2 x^{n-1} (x-1)^2 \\ \Leftrightarrow & (x-1)^2 (x^{n-1} + x^{n-2} + \dots + 1 - nx^{\frac{n-1}{2}})(x^{n-1} + x^{n-2} + \dots + 1 + nx^{\frac{n-1}{2}}) \geq 0. \end{aligned}$$

It suffices to show $x^{n-1} + x^{n-2} + \dots + 1 \geq nx^{\frac{n-1}{2}}$. This holds by the AM-GM inequality. Equality holds when $x = 1$ or $n = 1$. $\square$

Problem 106 (Mediterranean MO 2016 Problem 1). Given $\triangle ABC$, define the following points.

- D is the intersection point of the internal angle bisector of $\angle BAC$ and the side BC
- E is the intersection point of the tangent at A to the circumcircle of $\triangle ABC$ and the line BC
- F is the point on the A -altitude of $\triangle ABC$ such that $DF \perp AE$
- M is the midpoint of AB
- P is the intersection point of AB and EF
- Q is the intersection point of AD and EM
- O is the circumcentre of $\triangle ABC$

Prove that $PQ \perp AO$.



Solution. It is well-known that $EA = ED$ (for example, $\angle EAD = \angle EAB + \angle BAD = \angle ACB + \angle DAC = \angle ADE$). Since $AF \perp DE$ and $DF \perp AE$, F is the orthocentre of $\triangle ADE$, and so $EF \perp AD$. Thus, EF is the perpendicular bisector of AD . By Menelaus' theorem, we have

$$\frac{DQ}{QA} \times \frac{AM}{MB} \times \frac{BE}{ED} = 1.$$

This implies $\frac{DQ}{QA} = \frac{DE}{EB} = \frac{EA}{EB} = \frac{PA}{PB}$, where the last equality follows from the angle bisector theorem. Let X be the intersection point of DP and AE . As EP is the perpendicular bisector of AD , we have $\frac{PA}{PB} = \frac{PD}{PX}$ by symmetry. Combining these, we obtain $\frac{DQ}{QA} = \frac{DP}{PX}$. This implies $PQ \parallel XA$, which means $PQ \perp AO$ as $\angle OAE = 90^\circ$. $\square$

Problem 107 (Thailand MO 2021 Day 1 Problem 5). Find all prime numbers p and all odd positive integers m and n such that $m^4 + 4^n p^4$ divides $m^2(m^4 - 4^n p^4)$.

Solution. From $m^4 + 4^n p^4 \mid m^2(m^4 - 4^n p^4)$, we obtain $m^4 + 4^n p^4 \mid m^2(2 \cdot 4^n p^4)$. As $m^4 + 4^n p^4$ is odd, this implies

$$m^4 + 4^n p^4 \mid m^2 p^4. \quad (1)$$

If $p \nmid m$, then $(m^4 + 4^n p^4, p) = (m^4, p) = 1$. Therefore, (1) becomes $m^4 + 4^n p^4 \mid m^2$. But this is impossible since $0 < m^2 < m^4 + 4^n p^4$. Therefore, we may let $m = p\ell$ for some odd positive integer ℓ . Then (1) can be simplified to

$$\ell^4 + 4^n \mid p^2 \ell^2.$$

In addition, as $(\ell^4 + 4^n, \ell) = (4^n, \ell) = 1$, this can be further reduced to

$$\ell^4 + 4^n \mid p^2.$$

It follows that $\ell^4 + 4^n = p, p^2$ (note that $\ell^4 + 4^n > 1$).

We write $n = 2d - 1$ and $\ell^4 + 4^{2d-1} = p^s$ where $s = 1, 2$. Note that we can factorize the left-hand side so that

$$(\ell^2 + 2^{2d-1} - 2^d \ell)(\ell^2 + 2^{2d-1} + 2^d \ell) = p^s.$$

As the first factor is smaller, and $s = 1, 2$, we must have $\ell^2 + 2^{2d-1} - 2^d \ell = 1$ and $\ell^2 + 2^{2d-1} + 2^d \ell = p^s$. Note that

$$1 = \ell^2 + 2^{2d-1} - 2^d \ell = (\ell - 2^{d-1})^2 + 2^{2d-2}.$$

This forces $d = 1$ and $\ell = 1$. Then it is easy to obtain $p = 5$ and $s = 1$. Thus, $(m, n, p) = (5, 1, 5)$. One checks that $5^4 + 4 \cdot 5^4 \mid 5^2(5^4 - 4 \cdot 5^4)$, so this is the only solution. $\square$

Problem 108 (Serbia MO 2020 Problem 6). Let k and n be positive integers. We play a game on an infinite row of boxes as follows. Initially, we distribute n balls to the boxes in some ways, where each box may contain more than 1 ball. In each operation, we are allowed to make one of the following moves:

- We choose two adjacent nonempty boxes, and transfer all the balls from one box to the other.
- We choose a box with at least 2 balls. Then we transfer one ball to the k th box on the left, and transfer one ball to the k th box on the right.

The game ends when we cannot make any move.

- Prove that the game must end if $n \leq k + 1$.
- Determine all k such that there exists an n for which the game may never come to an end.

Solution.

- Claim.** If $k = 1$, and we are only allowed to use the second move, then the game must end for every n .

Proof. Assign the values $\dots, 2^2, 2, 1, 2, 2^2, \dots$ to the boxes. The value of a ball is the value of the box it lies in. Clearly, the total value of the balls are strictly increasing. Therefore, if the game never comes to an end, the total value must be unbounded. As the number of balls is finite, there must be some balls that move indefinitely to the right (WLOG).

When 2 balls in box $j + 1$ are distributed to box j and box $j + 2$, there is always 1 ball staying in either box $j + 1$ or box $j + 2$ forever. Thus, by partitioning all boxes into consecutive pairs of boxes, we see that there are more and more pairs which contain at least one ball. This is impossible as n is fixed. This proves the claim. $\square$

We now work on the original question. It suffices to consider $n = k + 1$, and we suppose on the contrary that the game never comes to an end. We label all boxes by the integers in the natural order. We put all boxes into k groups according to their labels modulo k .

If a group does not contain any ball at some moment, then it is clear that this group will never contain any ball afterwards. Therefore, we can partition all boxes into blocks of length $k - 1$ such that two adjacent blocks are separated by a box which will never contain any ball. We now replace the blocks by some large boxes. Note that the first move is only carried out within the same block, and we cannot always use the first move (as each time one more box is made empty). On the other hand, the second move is essentially to move two balls from a block to its two adjacent blocks. Therefore, we are in the situation of the Claim with the use of large boxes. By the claim, we are done.

Thus, we may assume all groups always contain at least one ball. As there are $k + 1$ balls and k groups, exactly one group consists of 2 balls and all other groups consist of 1 ball.

- If we use the second move, the two balls in the same box (hence the same group) are put in two different boxes (but in the same group). So we have to use the first move in the next step as there are no more boxes with two balls.
- If we use the first move, then the two boxes involved must contain one ball each (or otherwise there are at least 3 balls in a group afterwards). Afterwards, the two boxes contain 2 and 0 balls respectively. Now we cannot use the first move again or otherwise there is a group of at least 3 balls, or two groups of 2 balls. Therefore, we have to use the second move in the next step.

This shows we have to use the two moves alternately. Also, in view of the above discussion, we may identify a special ball which is transferred in every operation. We may assume the special ball visits all other balls infinitely often, or otherwise those other balls can be removed (and one group no longer contains any ball and we are done from above). In particular, the leftmost ball is always visited by the special ball. This shows the leftmost ball should move to the left indefinitely. Similarly, the rightmost ball should move to the right indefinitely.

On the other hand, if there are m balls (while k is the same as above) and we label the leftmost box containing a ball by 0, then we can show by induction that the leftmost ball cannot move beyond $-f(m)$ and the rightmost ball cannot move beyond $f(m)$ for some increasing function $f(m)$. Indeed, the base case $m = 2$ is obvious as at most two operations can be carried out.

For the inductive step (case m), WLOG assume the leftmost ball can move to the left indefinitely. It is not hard to see that the rightmost ball always lies in a box with a positive label (possibly except when there are two balls in box 0, but in the next move there must be a ball in box k). Therefore, eventually the leftmost and the rightmost balls are more than $(m - 1)f(m - 1)$ boxes apart. By the pigeonhole principle, we can divide the balls into two parts which are separated by more than $f(m - 1)$ boxes in between. We may focus on the part with the special ball. Since there are at most $m - 1$ balls in this part, by the inductive hypothesis, we see that the balls in this part can never be moved to the other part, including the special ball. This is impossible from above. This completes the inductive step, and yields a contradiction to the above observation. So we are done.

- (b) All values of k are possible. Consider $5k$ balls which are distributed evenly in $5k$ consecutive boxes with labels $1, 2, \dots, 5k$. Initially, the sequence of the number of balls in the boxes is $\underbrace{11 \dots 1}_{5k}$. We perform the following moves.

- Transfer the ball in box 1 to box 2, transfer the balls in box 2 to box 3, $\dots$, transfer the balls in box k to box $k + 1$. This ends up with the configuration $\underbrace{00 \dots 0}_k (k + 1) \underbrace{11 \dots 1}_{4k - 1}$.
- Transfer two balls from box $k + 1$ to box 1 and box $2k + 1$, transfer the balls in box $k + 1$ to box $k + 2$, transfer two balls from box $k + 2$ to box 2 and box $2k + 2$, transfer the balls in box $k + 2$ to box $k + 3$, $\dots$, transfer two balls from box $2k$ to box k and box $3k$. This ends up with the configuration $\underbrace{11 \dots 1}_k \underbrace{00 \dots 0}_k \underbrace{22 \dots 2}_k \underbrace{11 \dots 1}_{2k}$.
- Apply the above moves to the boxes in the right half in a symmetric way (starting from the rightmost box). This ends up with the configuration $\underbrace{11 \dots 1}_k \underbrace{00 \dots 0}_k \underbrace{33 \dots 3}_k \underbrace{00 \dots 0}_k \underbrace{11 \dots 1}_k$.
- Transfer two balls from box $2k + 1$ to box $k + 1$ and box $3k + 1$, transfer two balls from box $2k + 2$ to box $k + 2$ and box $3k + 2$, $\dots$, transfer two balls from box $3k$ to box $2k$ and box $4k$. This ends up with the initial configuration $\underbrace{11 \dots 1}_{5k}$.

Thus, we can repeat the same operations indefinitely without ending the game.

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